Final Project for CS 439

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Abstract

The abstract/summary goes here. I added some placeholders for the sections below.
https://github.com/craighymo/DataScienceProject

1 Introduction

To what extent can movie metadata and financial features explain or predict IMDb movie scores?

2 Related Work

Not sure what goes here but we'll figure it out.

3 Methodology

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ALL THE TECHNICAL STUFF IN HERE (but not really results)

preprocessing:
- renamed cols:
    names to title
    date_x to release_date
    budget_x to budget
    orig_lang to language

- parsed release dates and extracted release_year
- normalized text/categorical values and replaced missing text with "Unknown"
- kept only released movies with valid score, budget, and revenue
- grouped rare categorical values into "Other" to prevent rarer (random ass) languages and countr

feature engineering:
- profit = revenue - budget
- ROI = profit / budget
- roi_capped clips extreme ROI values using 1st and  99th percentiles.
- log_budget and log_revenue reduce skew.
- log_profit_shifted handles negative profit by shifting before log-transforming.

models
- mean score baseline
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40th Conference on Neural Information Processing Systems (NeurIPS 2026).

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- baseline linear regression (uses only budget and revenue)
- improved linear regression (uses engineered numeric features and one-hot encoded categorical fe
- majority classification baseline: always predicts the most common score category.
- logistic regression classification: predicts low/medium/high score quantile categories
    - initially had fixed thresholds for low/med/high but was imbalanced, medium group was huge
- K-Means + PCA: clusters movies using metadata/financial features, then visualizes clusters in t

preprocessing pipelines:
- numeric features
- categorical features
- Train/test split: 80/20 for supervised models
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4 Experiments & Results

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RESULTS (we'll show our plots and data here (btw that's in
- EDA
    - discuss RMSE, MAE, and R^2
- regression
- classification
    - logistic regression beats the majority-class baseline.
    -  model performs better on low/high categories than medium (hardest to separate).
    - (maybe we can fix this but I already messed around a bunch)

- coefficient interpretation
- clustering
    - K-Means, PCA
    - explorative NOT predictive
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5 Conclusion

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- models found some predictive signal in metadata and financial features
- improved regression beat both baselines
- logistic regression beat the majority-class   baseline for score categories.
- K-Means/PCA revealed meaningful movie groupings
- prediction is moderate tho: movie ratings are influenced by subjective factors not fully repres
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